Section 10:

Redshift Datawarehouse

Amazon Redshift

Amazon Redshift

- Fully managed, highly available, cost-effective petabyte-scale data warehouse service in the cloud.
- ANSI SQL compatible relational database.
- Can be accessed using Amazon Redshift query editor v2 or any other Business Intelligence (BI) tool.

Amazon Redshift

- Handles large amounts of data.
- Supports multiple data sources, including

CSV

JSON

Apache Parquet

Amazon Redshift

Features

- Scalable
- Columnar Storage
- Massively Parallel Processing (MPP)
- Integration with Other AWS Services
- Advanced Compression
- Security Features

Amazon Redshift

Use cases include

- Data Warehousing
- Business Intelligence
- Analytics
- Log Analysis
- IoT Data Processing
- Real-Time Dashboards

Amazon Redshift Cluster

Part One

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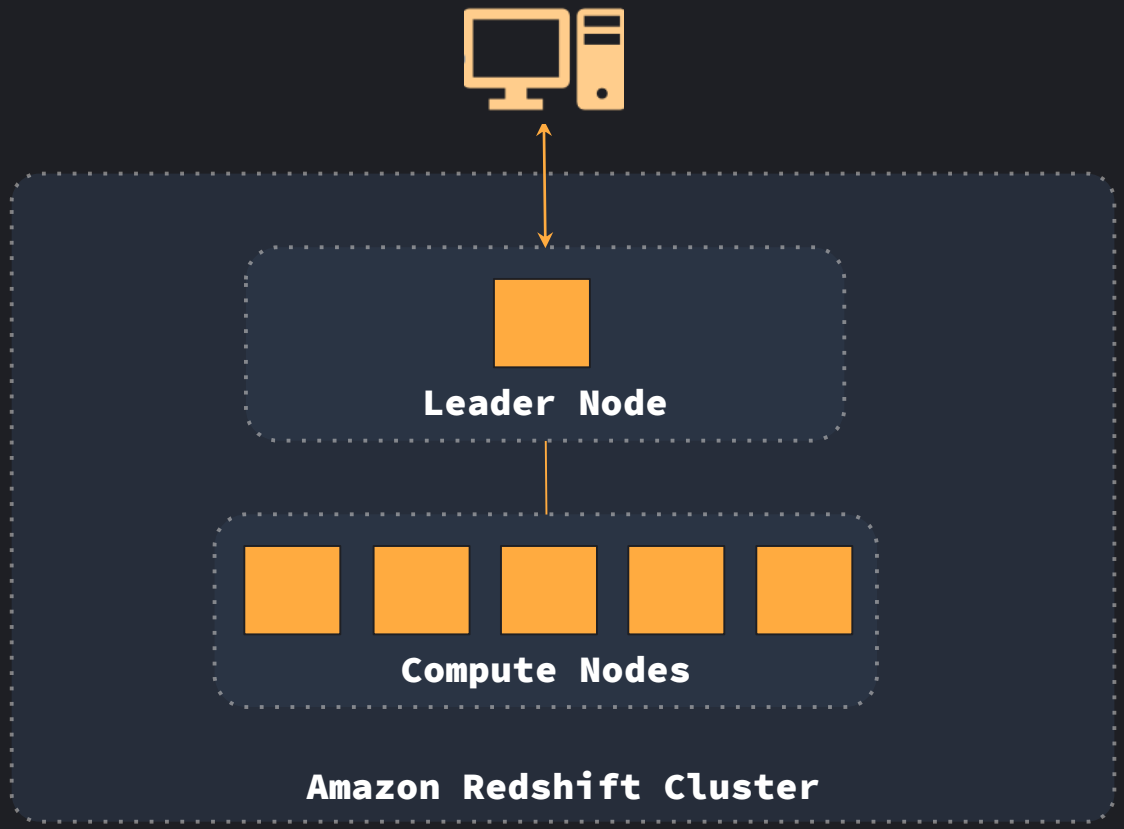
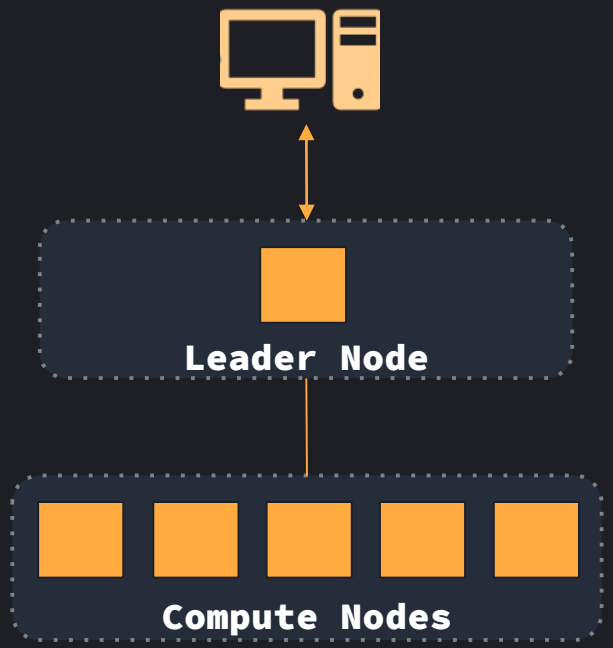
Amazon Redshift Cluster

- Is the core infrastructure component of Redshift.
- Runs an Amazon Redshift engine and contains one or more databases.
- Executes workloads coming from external data apps.
- Uses replication and continuous backups.
- Automatically recovers from failures.

Amazon Redshift Cluster



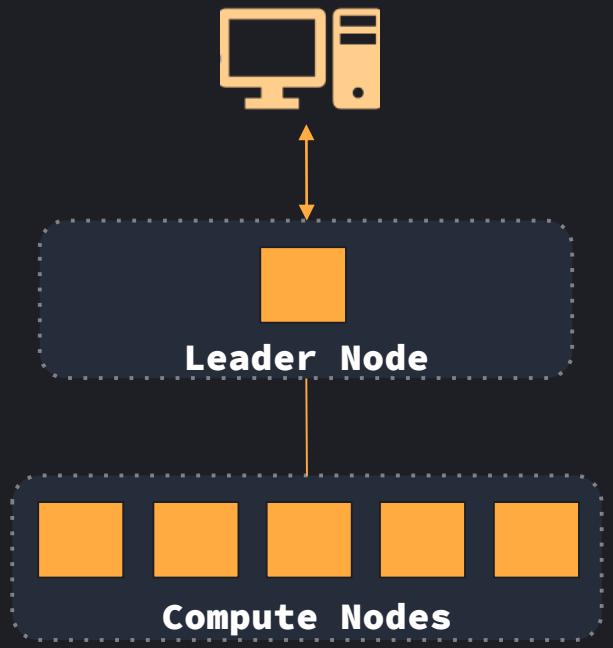
Amazon Redshift Cluster



Leader Node

- Coordinates two or more compute nodes.
- Aggregates results from compute nodes.
- Develops execution plan.
- Assigns a portion of the data to each compute node.

Amazon Redshift Cluster



Compute Nodes

- Has CPU, memory, and disk storage.
- Run the query execution plans.
- Transmit data among themselves.
- Capacity can be increased by increasing the number of nodes, upgrading the node type, or both.
- Use **provisioned cluster** or **Redshift Serverless**

Redshift Managed Storage

Redshift Managed Storage (RMS)

- Uses large, high-performance solid-state drives (SSDs) for fast local storage.
- Uses S3 for longer-term durable storage.
- Pricing is the same for RMS regardless of whether the data resides in high-performance SSDs or in S3.

RA3 and DC2 Node types

RA3 and DC2 Node types

RA3 nodes

- Uses Redshift Managed Storage (RMS).
- Separates compute and storage.
- Scale and pay for compute and managed storage independently.
- Supports multi Availability Zone (AZ)

RA3 and DC2 Node types

DC2 nodes

- Local SSD storage included.
- DC2 nodes store your data locally for high performance.
- Available on single Availability Zone (AZ) only.
- Recommend by AWS For datasets under 1TB (compressed).
- You can add more compute nodes to increase the storage capacity of the cluster.

Amazon Redshift Cluster

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Resizing clusters

Used to change the node type or the number of nodes in the cluster

Node Slices:

- Compute nodes are split into slices
- Each handling a part of the workload.
- Leader node distributes data & tasks to the slices for parallel processing

Elastic resize

- Adjust number of nodes without disruption or restart (by redistributing data slices across nodes)
- Completes quickly.

Can be used to …

- Add or remove nodes from your cluster.
- Change Node Type: From DS2 to RA3

Amazon Redshift Cluster

Classic resize

- Similar to elastic resize.
- Takes more time to complete
- Useful in cases were elastic resize is not applicable

Amazon Redshift snapshots

Amazon Redshift snapshots

- Snapshots are point-in-time backups of a cluster.

- Snapshots are stored internally on S3.

- Snapshot can be taken manually or automatically.

Amazon Redshift snapshot

Automated Snapshot

- Incremental snapshots are taken automatically.
- Is enabled by default.
- By Default Snapshot is taken

Every Eight Hours

Every 5 GB per node of data changes

- Default retention period is one day.

Amazon Redshift snapshots

Manual Snapshots

- Can be taken any time.
- By default, manual snapshots are retained indefinitely.
- You can specify the retention period when you create a manual snapshot, or you can change the retention period by modifying the snapshot.

Sharing data across AWS Regions

Sharing data across AWS Regions

- You can share data across AWS Regions without the need to copy data manually.
- Sharing can be done without Amazon S3 as a medium.
- With cross-Region data sharing, you can share data across clusters in the same AWS account, or in different AWS accounts even when the clusters are in different Regions.

Distribution Styles

Distribution Styles

- Clusters store data across compute nodes.
- Use Distribution Keys for that distribution
- Distribution Styles determine where data is stored (compute node)
- Distribute the workload uniformly among the nodes in the cluster
- Minimize data movement during query execution.

Distribution Styles

KEY distribution

- Rows are distributed according to the values in one column.
- The leader node places matching values on the same node slice.
- Useful for tables that participate frequently in joins

Distribution Styles

ALL distribution

- A copy of the entire table is distributed to every node.
- Multiplies the storage required by the number of nodes in the cluster.
- Is appropriate only for relatively small and slowly changing tables.
- Faster query operations

Distribution Styles

EVEN distribution

- Leader node distributes the rows across the slices. (regardless of the values in any particular column)
- Appropriate when a table doesn't participate in joins.
- Appropriate when there isn't a clear choice between KEY distribution and ALL distribution.

Distribution Styles

AUTO distribution

- Redshift assigns an optimal distribution style based on the size of the table data.
- Is the default distribution Style
- How does it work

- The change in distribution style occurs in the background with minimal impact to user queries.

Vacuum and Workload Management

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Vacuum

Vacuum

- It **Re-sorts rows** and **reclaims space** in either a specified table or all tables in the current database.
- Redshift automatically sorts data and runs VACUUM DELETE in the background.
- By default, VACUUM skips sorting for tables where > 95% already sorted.
- Users can access tables while they are being vacuumed.
- Redshift automatically performs a DELETE ONLY vacuum in the background.

Vacuum

Command Syntax

```
VACUUM [ FULL | SORT ONLY | DELETE ONLY | REINDEX | RECLUSTER ]
[ [ table_name ] [ TO threshold PERCENT ] [ BOOST ] ]
```

- FULL : Sorts the specified table and reclaims disk space.
- SORT ONLY : Sorts the specified table (or all tables) without reclaiming space
- DELETE ONLY : Reclaims disk space only.
- REINDEX : Rebuilds the indexes on the tables. Useful for tables using Interleaved Sort Keys.
- RECLUSTER : Sorts the portions of the table that are unsorted.

Vacuum

- TO *threshold* PERCENT :
  Specifies a threshold above which VACUUM skips sort phase & reclaiming space in delete phase.

- BOOST :
  Runs the VACUUM command with additional resources as they are available.

Redshift Integration

AWS Redshift Integration

S3

EMR

Lambda

DMS

EC2

Data pipeline

MSK

KDS

DynamoDB

Copy command

```
copy favoritemovies
from 'dynamodb://Movies'
iam_role'arn:aws:iam::0123456789012:role/MyRedshiftRole;
```

- Is used to load large amount of data from outside of redshift.
- Uses the Amazon Redshift the Massively Parallel Processing (MPP).
- Uses optimal compression scheme.
- Supported compression include

Gzip | IZop | bzip

Moving data between S3 and Redshift

Loading Data from S3 to Redshift

- You can add data to your Amazon Redshift tables either by using an INSERT or COPY command.
- loading data with COPY
  - Is faster.
  - Decrypts data as it is loaded from S3.
  - Requires a manifest file and IAM role.

Unloading Data to S3 from Redshift

- The UNLOAD command allows you to export the results of a query to Amazon S3.
- It supports exporting data in various formats, including

CSV | Apache Parquet | ORC

Moving data between S3 and Redshift

Redshift Auto Copy from Amazon S3

- Automatically copies data from an Amazon S3 bucket into an Amazon Redshift.

Enhanced VPC routing

- Enables you to route the network traffic through a VPC instead of the internet.

Amazon Aurora zero-ETL integration

- Allows changes made to the Amazon Aurora RDBMS database to be replicated in the Amazon Redshift database seconds after the Aurora updates.
- Eliminates the need for custom data pipelines
- Enables customers to analyze petabytes of transactional data in near real time.

Data Transformation using Amazon Redshift

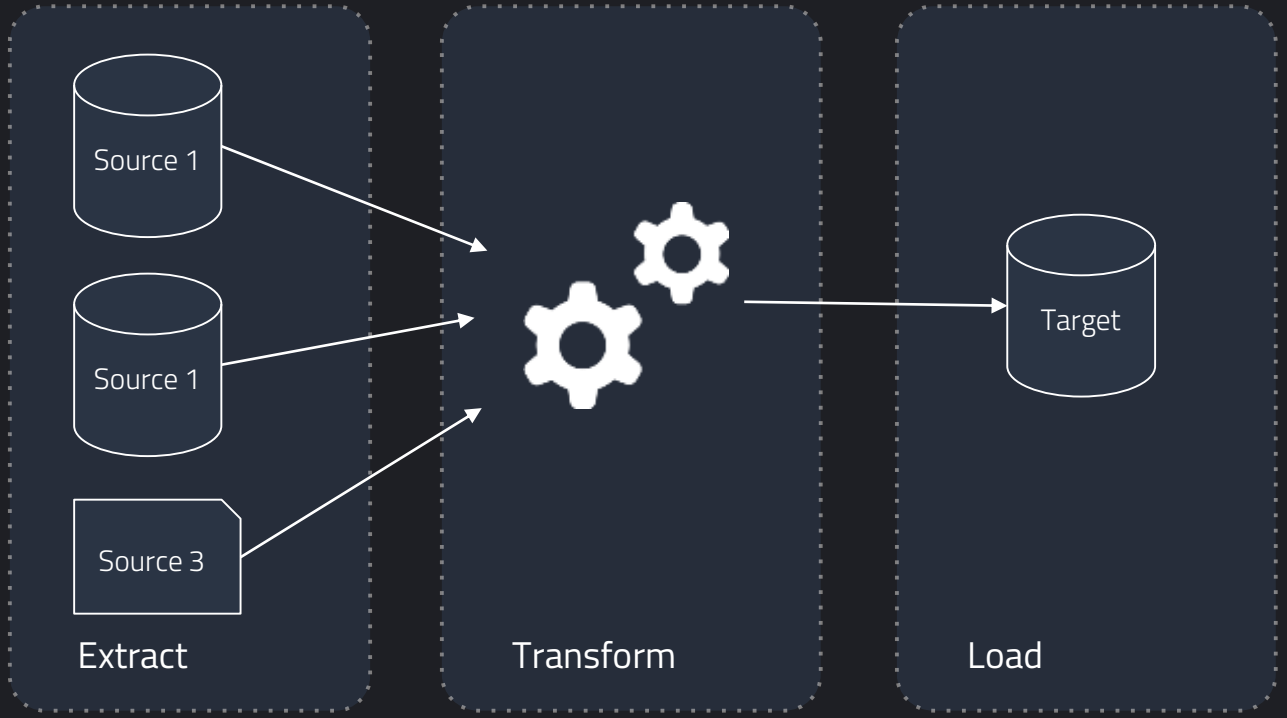
Data Transformation using Amazon Redshift

Design patterns when moving data from source systems to a data warehouse

- Extract Transform Load (ETL)
- Extract Load Transform (ELT)

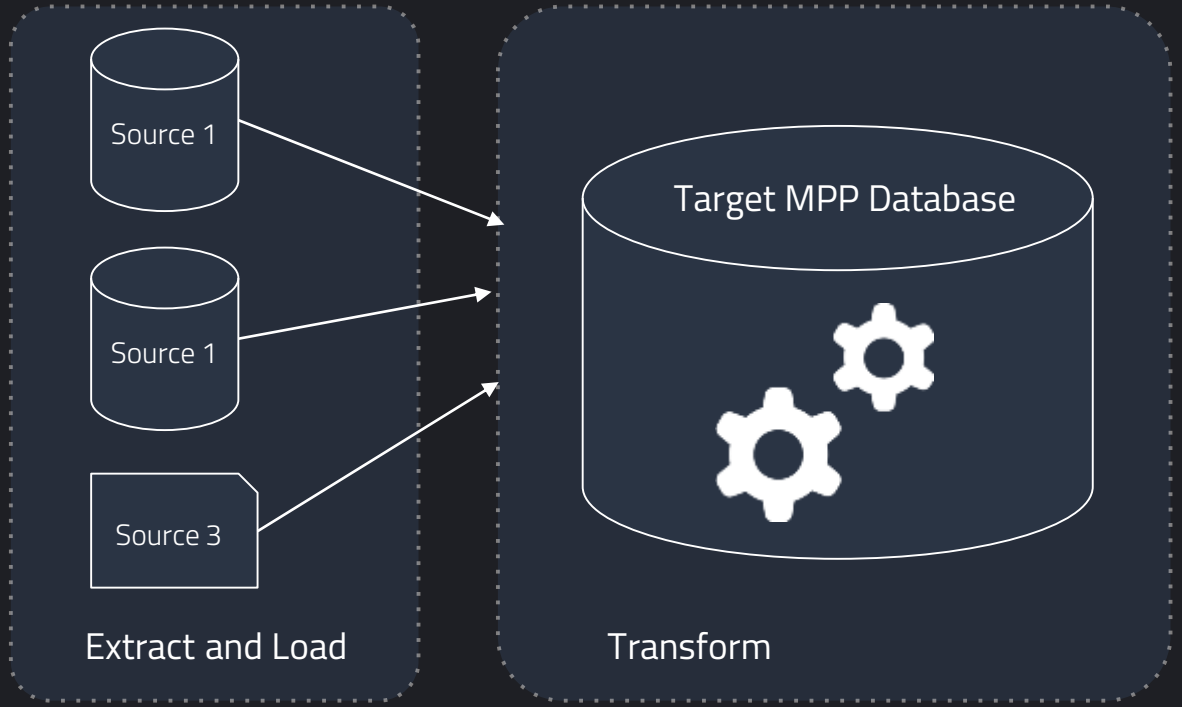
Data Transformation using Amazon Redshift

Extract Transform Load (ETL)



Data Transformation using Amazon Redshift

Extract Load Transform (ELT)



Data Transformation using Amazon Redshift

- Amazon Redshift provides a functionality to process all your data in one place with its in-database transformation (ELT) capabilities.
  - SQL Transformations
  - Stored Procedures
  - User-Defined Functions (UDFs)

Data Transformation using Amazon Redshift

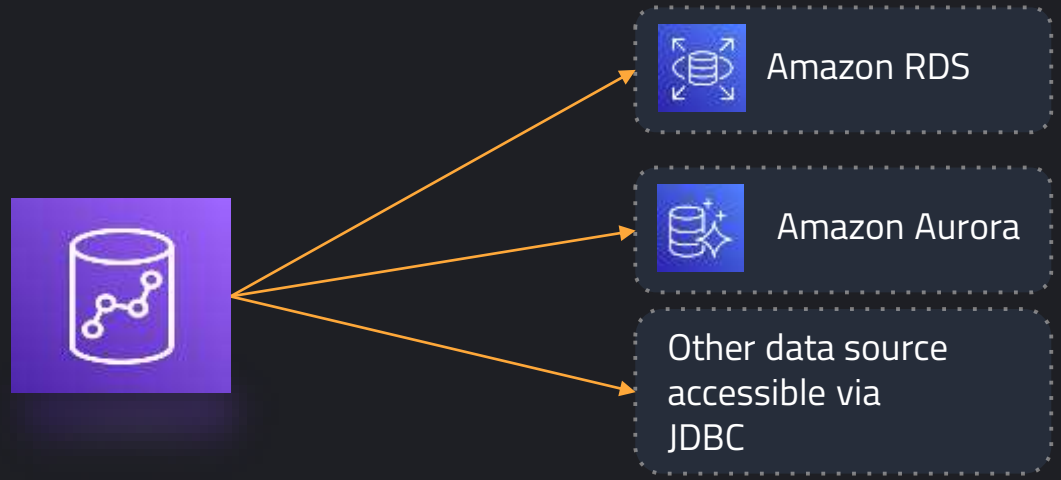
- Amazon Redshift can connect to an ETL platform of using JDBC and ODBC.
- Popular ETL platforms that integrate with Amazon Redshift include third-party tools like
  - Informatica
  - Matillion
  - dbt
  - AWS-native tools like “AWS Glue”.

Amazon Redshift Federated Queries

Amazon Redshift Federated Queries

- Used to combine and analyze data across different data sources.
- Reduces data transmission while optimizing efficiency.
- Eliminates the necessity of ETL pipelines.

Amazon Redshift Federated Queries



Amazon Redshift Federated Queries

- Uses external schema definitions.
- Can be used to incorporate live data.
- Distributes part of the computation for federated queries directly into the remote operational databases.
- Uses parallel processing capacity.

Materialized Views

Materialized Views

- Contains a precomputed result set.
- You can issue SELECT statements to query it.
- Redshift returns the precomputed results from the materialized view.

Materialized Views

- Query results are returned much faster.
- Useful for speeding up queries that are predictable and repeated.
- You can define a materialized view in terms of other materialized views.

```
CREATE MATERIALIZED VIEW mv_name
AUTO REFRESH { YES | NO }
AS query
```

Materialized Views

Streaming Ingestion

- Ingest data from streaming sources

Steps for creating a streaming ingestion

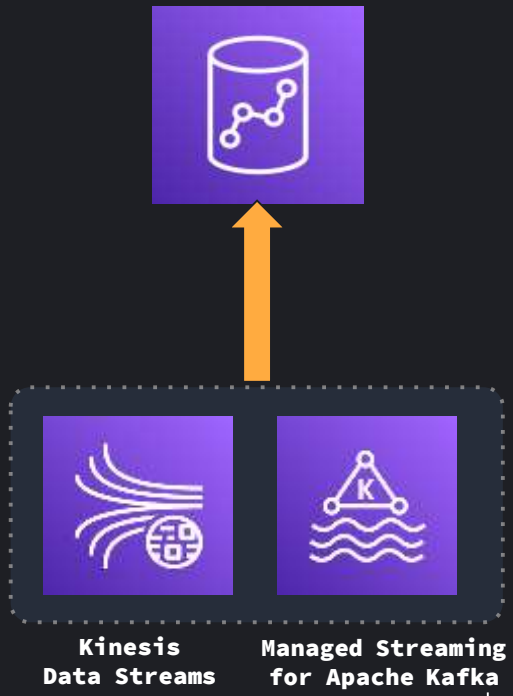
1. Create an external schema that maps to the streaming data source.

```
CREATE EXTERNAL SCHEMA evdata FROM KINESIS
IAM_ROLE 'arn:aws:iam::0123456789:role/redshift-streaming-role';
```

2. Create a materialized view to consume the data

```
CREATE MATERIALIZED VIEW view_name
AUTO REFRESH YES
AS SELECT … ;
```

Turn on Auto-refresh



Materialized Views

Updating materialized view

- Automatic refresh option to refresh materialized views when base tables of materialized views are updated.
- Auto refresh operation runs at a time when cluster resources are available to minimize disruptions to other workloads.
- Specify AUTO REFRESH in the materialized view definition to enable it.

Materialized Views

- Scheduler API and console integration.
- You can also manually refresh any materialized views using the REFRESH MATERIALIZED VIEW statement.

Materialized Views

Commands used to create and manage materialized views

- CREATE MATERIALIZED VIEW
- ALTER MATERIALIZED VIEW
- REFRESH MATERIALIZED VIEW
- DROP MATERIALIZED VIEW

Amazon Redshift Spectrum

Amazon Redshift Spectrum

- Enables you to run complex SQL queries directly against data stored in Amazon S3.

- You define external tables in Redshift cluster that reference the data files stored in S3.

Amazon Redshift Spectrum

- Handles the execution of queries by retrieving only the necessary data from S3.
- Supports different data formats, Gzip and Snappy compression.
- Resides on dedicated Amazon Redshift servers.
- Pushes many compute-intensive tasks down to the Redshift Spectrum layer.

Amazon Redshift Spectrum

- Scales intelligently.
- Redshift Spectrum tables can be created by defining the structure for your files and registering them as tables in an external data catalog.
- The external data catalog can be

AWS Glue

Amazon Athena

Apache Hive metastore

Amazon Redshift Spectrum

- Changes to the external data catalog are immediately available.
- You can partition the external tables on one or more columns.
- Redshift Spectrum tables can be queried and joined just as any other Amazon Redshift table.

Amazon Redshift Spectrum

Amazon Redshift Spectrum considerations

- Redshift cluster and the S3 bucket must be in the same AWS Region.
- Redshift Spectrum doesn't support enhanced VPC routing with provisioned clusters.
- Supports Amazon S3 access point aliases.
- Redshift Spectrum doesn't support VPC with Amazon S3 access point aliases

Amazon Redshift Spectrum

Amazon Redshift Spectrum considerations (continued)

- You can't perform update or delete operations on external tables.
- To create a new external table in the specified schema, you can use CREATE EXTERNAL TABLE.
- To insert the results of a SELECT query into existing external tables on external catalogs, you can use INSERT (external table).
- Unless you are using an AWS Glue Data Catalog that is enabled for AWS Lake Formation, you can't control user permissions on an external table.

Amazon Redshift Spectrum

Amazon Redshift Spectrum considerations (continued)

- To run Redshift Spectrum queries, the database user must have permission to create temporary tables in the database.

- Redshift Spectrum doesn't support Amazon EMR with Kerberos.

System Tables and Views

Redshift System Tables and Views

System Table

- Contain information about how the system is functioning.

System View

- Views to organize information from the system tables.

Redshift System Tables and Views

- Queried the same way like any other database tables.
- Some system tables can only be used by AWS staff for diagnostic purposes.

Redshift System Tables and Views

Types of system tables and views

- SVV views

  Details on database objects;
  SVV_ALL_TABLES see all tables
  SVV_ALL_COLUMNS see a union of columns

- SYS views

  Monitor query and workload performance in clusters;
  SYS_QUERY_HISTORY see details of user queries

- STL views

  Generated from system logs for historical records;
  STL_ALERT_EVENT_LOG identify opportunities to improve query performance
  STL_VACUUM statistics for tables that have been vacuumed

- STV tables

  Snapshots of the current system data;
  STV_EXEC_STATE information about queries & query steps actively running

Redshift System Tables and Views

Types of system tables and views

- SVCS views — Details about queries on both the main and concurrency scaling clusters; **SVCS_QUERY_SUMMARY** general information about the execution of a query
- SVL views — Contain references to STL tables & logs for more detailed information; **SVL_USER_INFO** data about Amazon Redshift database users

Redshift Data API

Redshift Data API

- It is a lightweight, HTTPS-based API that is used for running queries against Amazon Redshift.
  - Lambda
  - SageMaker notebooks
  - Other web-based applications

Serverless, event-driven architecture	Web applications	Third-party services

- Can be used to run SQL queries asynchronously.
- Can be used as an alternative to using JDBC or ODBC drivers.
- No need to manage and maintain WebSocket or JDBC connections.
- Can be setup with very little operational overhead (very easy setup!)

Redshift Data API

- Serverless Execution
- Asynchronous Processing
- Direct Integration with AWS Services (e.g. Lambda / SageMaker)
- Simplified Management

Redshift Data API

- Maximum duration of a query is 24 hours.
- Maximum number of active queries per Amazon Redshift cluster is 200.
- Maximum query result size (after compression) is 100 MB.
- Maximum retention time for query results is 24 hours.

Redshift Data API

- Maximum query statement size is 100 KB.

- Is available to query single-node and multiple-node clusters of the following node types:

dc2.large | dc2.8xlarge | ra3.xlplus | ra3.4xlarge

ra3.16xlarge

Acess Control

- Authorize user/service by adding managed policy
  ⇒ *AmazonRedshiftDataFullAccess*

Monitoring Data API

- Monitoring of Data API events in EventBridge
- EventBridge routs data to targets such as AWS Lambda or SNS.
- Option to schedule Data API operations within EventBridge

Data Sharing

Data Sharing

- Used to securely share access to live data across

Clusters

Workgroups

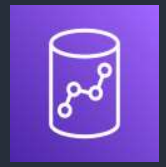
AWS accounts

AWS Regions

Availability Zones

- Data is live

Data Sharing



Producer Cluster

Datashare

- Tables
- Views
- User defined functions

Consumer Cluster

Data Sharing

Standard Datashares

AWS Data Exchange datashares

AWS Lake Formation-managed datashares

Data Sharing

- The consumer is charged for all compute and cross-region data transfer fees
- The producer is charged for the underlying storage of data
- The performance of the queries on shared data depends on the compute capacity of the consumer clusters.
- Data sharing continues to work when clusters are resized or when the producer cluster is paused.

Data Sharing

- Data sharing is supported for all provisioned ra3 cluster types and Amazon Redshift Serverless.
- For cross-account and cross-Region data sharing, both the producer and consumer clusters and serverless namespaces must be encrypted.
- You can only share SQL UDFs through datashares. Python and Lambda UDFs aren't supported.

Data Sharing

- Adding external schemas, tables, or late-binding views on external tables to datashares is not supported.

- Consumers can't add datashare objects to another datashare.

Workload Management (WLM)

Workload Management (WLM)

- Helps you manage query performance for competing queries
- Define how resources are allocated to different queries.
- Uses query queues
  ⇒ Route queries to the appropriate queues
- Enables users to flexibly manage priorities within workloads.
- You can create up to 8 queues.

Workload Management (WLM)

Automatic Work Load Management

- Redshift manages:
  - How many queries run concurrently
  - How much memory is allocated to each dispatched query.
- When complex queries ⇒ lower concurrency
- When lighter queries ⇒ higher concurrency

Workload Management (WLM)

Automatic Work Load Management

- You can configure the following for **each query queue**:

Queue	Configurable with different priorities and rules
User groups	assign a set of user groups to a queue
Query groups	assign a set of query groups to a queue
Priority	relative importance of queries
Concurrency scaling mode	automatically adds additional cluster capacity

Workload Management (WLM)

Manual Work Load Management

- Manage system performance by modifying WLM configuration.
- You can configure the amount of memory allocated to each queue.

⇒ Automatic usually higher throughput

⇒ Manual offers more control

Workload Management (WLM)

Short query acceleration (SQA)

- Prioritizes selected short-running queries over longer-running queries.
- Uses machine learning to predict the execution time of a query.
- Runs short-running queries in a dedicated space.

Workload Management (WLM)

Short query acceleration (SQA)

- CREATE TABLE AS (CTAS) statements and read-only queries, such as SELECT statements, are eligible for SQA.

Redshift Serverless

Redshift Serverless

- No Cluster Management.
- On demand scaling.
- Pay as you go.

Redshift Serverless

Node

- Automatically provisions and manages capacity for you.

Workload management and concurrency scaling

- Automatically manages resources efficiently and scales, based on workloads, within the thresholds of cost controls.

VS

Redshift provisioned

Node

- You build a cluster with node types that meet your cost and performance specifications

Workload management and concurrency scaling

- You enable concurrency scaling on your cluster to handle periods of heavy load.

Redshift Serverless

Port

- You can choose port from the port range of 5431-5455 or 8191-8215.

Resizing

- Not Applicable

Encryption

- Always encrypted with AWS KMS, with AWS managed or customer managed keys.

VS

Redshift provisioned

Port

- You can choose any port to connect

Resizing

- Cluster resize can be done to add nodes or remove nodes

Encryption

- Can be encrypted with AWS KMS (with AWS managed or customer managed keys), or unencrypted.

Redshift ML

Redshift ML

- Allows users to create, train, and apply machine learning models using just SQL.
- Users can utilize SQL commands to create and manage machine learning models, which are then trained using data stored in Redshift.
- Amazon Redshift ML enables you to train models with one single SQL CREATE MODEL command.

Redshift ML

- Lets users create predictions without the need to move data out of Amazon Redshift.
- Users can create, train, and deploy machine learning models directly within the Redshift environment.
- Useful for users that doesn't have expertise in machine learning, tools, languages, algorithms, and APIs.

Redshift ML

- Redshift ML supports common machine learning algorithms and tasks, such as

Binary classification

Multiclass classification

Regression

- It Automatically finds the best model using Amazon SageMaker Autopilot.

- Can use the massively parallel processing capabilities of Amazon Redshift.

Security in Amazon Redshift

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Amazon Redshift Spectrum

Amazon Redshift provides these features to manage security

- Sign-in credentials
- Access management using IAM
- Virtual Private Cloud (VPC)
- Cluster encryption

Redshift Security

- **Cluster security groups**
  - **Default Lockdown:**
    When you provision an Amazon Redshift cluster, it is locked down by default so nobody has access to it.
  - **Inbound Access Control:**
    To grant other users inbound access to an Amazon Redshift cluster, you associate the cluster with a security group.
- **SSL connections** to secure data in transit
- **Load data encryption** to encrypt the data during the load process

Amazon Redshift Spectrum

- Data in transit
- Column-level access control
- Row-level security control

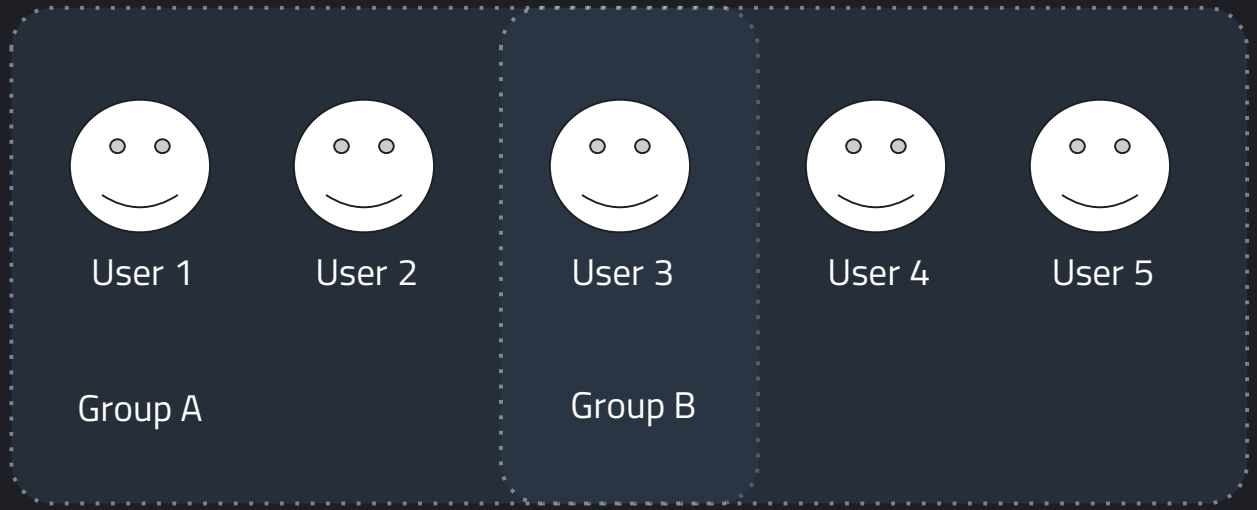
Access Control In Redshift

Access Control In Redshift

- You can either manage your users and groups within
  - Redshift
  - AWS IAM users
- The privileges to access specific objects are tightly coupled with the DB engine itself.

Access Control In Redshift

Manage your users and groups within Redshift



Access Control In Redshift

```sql
CREATE USER human_resource_1 PASSWORD 'xyzzy-1-XYZZY';
ALTER GROUP human_resource ADD USER human_resource_1;
```

```sql
CREATE USER operation_1 PASSWORD 'xyzzy-1-XYZZY';
ALTER GROUP operation ADD USER operation_1;
```

Access Control In Redshift

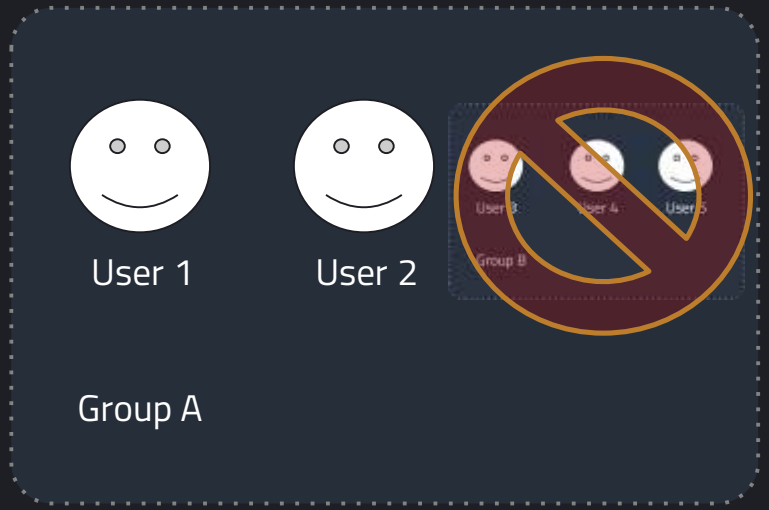
```
/* Creating the 2 groups */

CREATE GROUP humanresource;
CREATE GROUP operation;
```

```
/* Creating the 2 schemas with the data */

CREATE SCHEMA humanresource;
CREATE SCHEMA operation;
```

```
/* Give sales USAGE rights in schema, and read-only (SELECT) access to the
tables within the schema */

GRANT USAGE on SCHEMA humanresource TO GROUP humanresource;
GRANT SELECT ON ALL TABLES IN SCHEMA operation TO GROUP operation;
```

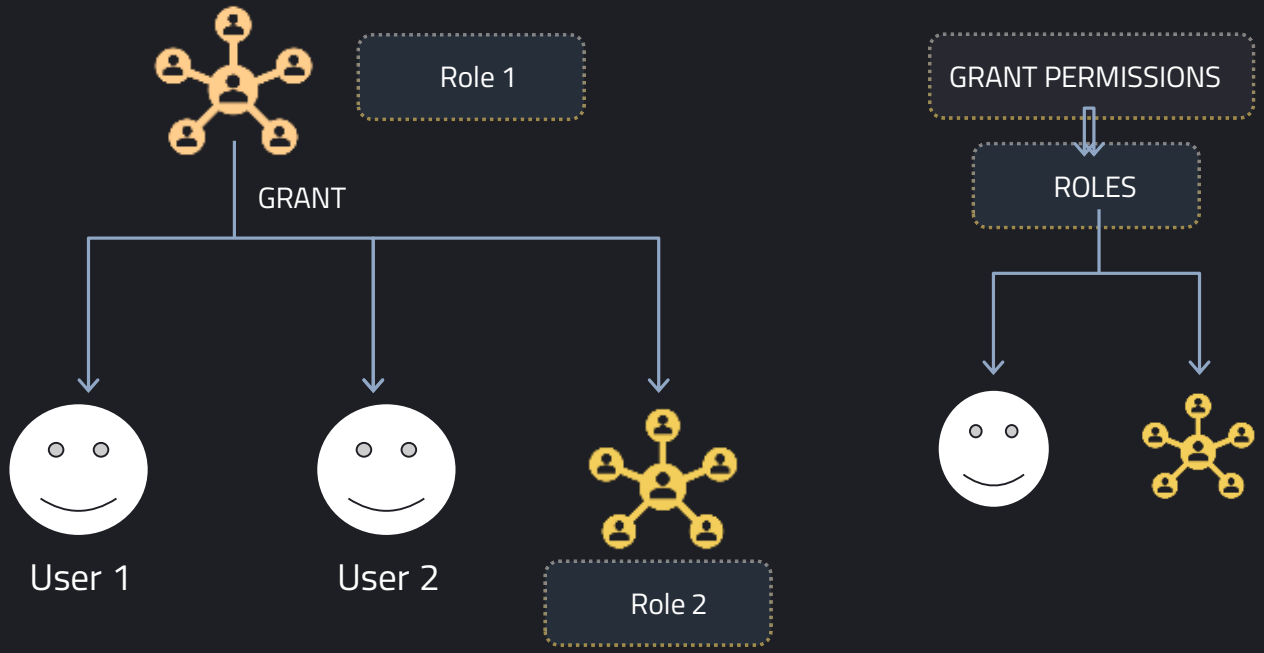
Access Control In Redshift

A group CANNOT contain another group



Access Control In Redshift

AWS introduced RBAC (Role-based access control)



Access Control In Redshift

```
/* Creating role*/

CREATE ROLE humanresource;
```

```
/* Grant role to user*/

GRANT ROLE humanresource TO user_1;
```

```
/* Give sales USAGE rights in schema, and read-only (SELECT) access to the
tables within the schema */

GRANT USAGE on SCHEMA humanresource TO ROLE humanresource;
GRANT SELECT ON ALL TABLES IN SCHEMA operation TO ROLE finance;
GRANT ROLE finance TO ROLE humanresource;
```

Redshift Fine-grained access control

Fine-grained Redshift access control

- Used to control and manage access permissions for various users or roles.
- Contains detailed access policies.

Fine-grained access control

```
/* Column-level access control*/

GRANT SELECT (column1, column2) ON tablename TO username;
```

Fine-grained access control

```sql
/* Row-level access control */

CREATE RLS POLICY view_own_warehouse_inventory
   WITH (warehouse_id INTEGER)
   USING (
       warehouse_id IN (
           SELECT managed_warehouse_id
           FROM warehouse_managers
           WHERE manager_username = current_user
       )
   );
```

item_id	item_name	quantity	warehouse_id
101	LED Light Bulb	120	1
102	Electric Drill	85	2
103	Hammer	75	1
104	Nails (Pack of 100)	150	3

manager_id	manager_username	managed_warehouse_id
1	jsmith	1
2	mbrown	2
3	lwilson	3

Fine-grained access control

```
/* Masking Policy */

CREATE MASKING POLICY mask_email
WITH (email VARCHAR(256))
USING ('***'::TEXT);
```

manager_id	manager_username	email
1	jsmith	***
2	mbrown	***
3	lwilson	***

Fine-grained access control

```
/* Create Masking Policy*/

CREATE MASKING POLICY mask_email
WITH (email VARCHAR(256))
USING ('***'::TEXT);
```

```
/* Attach Masking Policy*/

ATTACH MASKING POLICY mask_email
ON employee_data(email)
TO ROLE role_hr
PRIORITY 20;
```

manager_id	manager_username	email
1	jsmith	***
2	mbrown	***
3	lwilson	***

```
/* Detach Masking Policy*/

DETACH MASKING POLICY mask_email
ON employee_data(email)
FROM ROLE role_hr;
```

Redshift Fine-grained access control

Row level security in Redshift

- Users can only access specific rows.
- Rows have criteria that defines which role can access the specific item (row).

Access logging & monitoring in Redshift

- Failed and successful access attempts to Redshift data warehouses can be logged using the system table STL_CONNECTION_LOG.
- Audit logs are not enabled by default.